Case Study Report — Data Analyst 2025

Talent Match Intelligence System

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Executive Summary

Company X faces a critical challenge in succession planning: how to systematically identify which employees most closely match the profile of proven top performers. This project builds a **Talent Match Intelligence** system that answers that question at scale.

What was built:

1. **Success Pattern Discovery** — a data-driven analysis of 2,010 employees across 5 years of performance data, identifying what truly differentiates Rating-5 employees across competencies, psychometrics, behavioral assessments, and organizational context.
2. **SQL Matching Algorithm** — a 17-CTE parameterized query that computes how closely any employee matches a manager-defined benchmark profile, producing interpretable TV → TGV → Final Match Rate scores.
3. **AI-Powered Talent App** — a publicly deployed Streamlit dashboard that accepts runtime vacancy inputs, computes match scores dynamically, generates LLM-produced job profiles via OpenRouter, and visualizes insights through interactive charts.

Key outcome: Any manager can now define a vacancy, select 1–3 top performers as benchmark, and instantly receive a ranked list of 2,010 employees scored against that profile — with full transparency into which competencies, traits, and work preferences drive the match.

Section 1 — Success Pattern Discovery

1.1 Dataset Overview

Table	Rows	Notes
employees	2,010	Core entity
performance_yearly	10,050	5 years (2021–2025)
profiles_psych	2,010	One row per employee
papi_scores	40,200	20 scales × 2,010 employees
competencies_yearly	100,500	10 pillars × 5 years × 2,010
strengths	28,140	14 CliftonStrengths themes per employee

Data quality issues found: - Invalid ratings: 15 records with rating=0, 13 with rating=6, 7 with rating=99 → excluded from all analysis - Null psychometric data: IQ missing for 456 employees (22.7%), GTQ missing for 332 (16.5%) — these are department-level assessment gaps, not data errors - Invalid MBTI entries: some contained typos (e.g., "INFTJ") → cleaned to 16 valid types only

High performers (Rating=5): - 187 employees qualify as Rating=5 out of 1,975 valid (9.5%) - Distributed across all three directorates (Commercial, Technology, HR & Corp Affairs) - Average tenure: 4.3 years vs 4.2 years for others — tenure alone is not a differentiator

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1.2 Competency Execution — Strongest Signal

All 10 competency pillars show significant positive differences for R5 employees. The top performers score consistently higher across every dimension, not just selectively.

Pillar	Label	R5 Mean	Others Mean	Δ
SEA	Social Empathy & Awareness	5.03	2.92	**+2.10**
VCU	Value Creation for Users	5.01	3.17	**+1.83**
CSI	Commercial Savvy & Impact	5.04	3.22	**+1.82**
STO	Synergy & Team Orientation	4.96	3.15	**+1.81**
CEX	Curiosity & Experimentation	4.41	2.96	+1.44
QDD	Quality Delivery Discipline	4.56	3.16	+1.40
LIE	Lead, Inspire & Empower	4.47	3.18	+1.29
GDR	Growth Drive & Resilience	4.40	3.13	+1.27

Pillar	Label	R5 Mean	Others Mean	Δ
IDS	Insight & Decision Sharpness	4.40	3.14	+1.26
FTC	Forward Thinking & Clarity	4.39	3.17	+1.21

Insight: The top 4 differentiators — Social Empathy, Value Creation, Commercial Savvy, and Synergy — point to a success profile built around **people-orientation and business impact**, not just technical execution. R5 employees are not simply harder workers; they operate at the intersection of human connection and commercial results.

1.3 Work Preferences (PAPI) — Key Differentiators

PAPI differences are smaller in magnitude than competencies, but reveal a clear behavioral signature.

Positive for R5 (higher = more like top performers):

Scale	Label	R5	Others	Δ
Papi_P	Planning	5.32	4.96	**+0.35**
Papi_W	Need to Belong	5.10	4.98	+0.12
Papi_E	Empathy	5.16	5.06	+0.10
Papi_N	Need for Achievement	5.18	5.08	+0.10
Papi_F	Forward Planning	5.14	5.05	+0.09

Negative for R5 (lower = more like top performers):

Scale	Label	R5	Others	Δ
Papi_G	Leadership	4.61	4.97	**−0.35**
Papi_S	Social Harmony	4.73	5.02	−0.30
Papi_A	Attention to Detail	4.77	5.06	−0.29
Papi_T	Tenacity	4.73	4.98	−0.25

Insight: The counterintuitive finding — R5 employees score *lower* on the PAPI Leadership (G) scale — reveals that high performers at the current grade level drive results through **collaboration and planning discipline**, not positional authority. They are achievers who belong (Papi_W ↑) and plan (Papi_P ↑), not command-and-control leaders (Papi_G ↓).

1.4 Psychometric Profile

Measure	R5 Mean	Others Mean	Δ	Interpretation
Pauli	62.7	59.7	**+3.05**	Mental arithmetic stamina — strongest cognitive differentiator
GTQ	28.6	27.4	**+1.20**	General aptitude — moderate positive signal
IQ	109.0	109.6	−0.59	No meaningful difference
TIKI	5.5	5.5	+0.02	No meaningful difference
Faxtor	58.6	60.5	−1.89	Slightly lower in R5 (internal attention composite)

Insight: Sustained cognitive stamina (Pauli) and aptitude (GTQ) matter, but raw IQ does not differentiate top performers. This aligns with research showing that IQ predicts entry-level performance while sustained effort and structured reasoning (Pauli, GTQ) predict consistent excellence.

1.5 Behavioral Strengths

CliftonStrengths — Top themes in R5 (top-5 per employee):

Theme	Count	% of R5
Futuristic	37	19.8%
Restorative	37	19.8%
Intellection	35	18.7%
Self-Assurance	33	17.6%
Activator	30	16.0%
Strategic	30	16.0%
Learner	29	15.5%
Belief	28	15.0%
Positivity	28	15.0%
Analytical	27	14.4%

Pattern: R5 employees cluster around **future-oriented thinking** (Futuristic, Strategic, Forward Planning) combined with **self-driven problem solving** (Restorative, Activator, Self-Assurance). This maps directly to the competency profile: visionary problem-solvers who take initiative without waiting for permission.

DISC Type: No single DISC type dominates R5. The distribution is spread across CD (9.6%), SI (9.1%), CI (8.6%), DI (8.6%), DC (8.6%). High performance is not style-dependent — it's execution-dependent.

1.6 Contextual Factors

- **Tenure:** R5 avg 4.3 yrs vs 4.2 yrs — virtually identical. Tenure alone does not predict top performance.
- **Grade:** R5 employees exist at all three grade levels (III, IV, V), confirming the pattern is not grade-specific.
- **Directorate:** R5 representation is proportional across all three directorates.

1.7 Success Formula

Based on the above analysis, the weighted Success Formula is:

```
Final Match Rate =
    0.30 × Cognitive Ability Score
  + 0.20 × Work Preferences Score
  + 0.25 × Competency Execution Score
  + 0.15 × Behavioral Strengths Score
  + 0.10 × Contextual Fit Score
```

Weight Rationale:

TGV	Weight	Justification
Cognitive Ability	0.30	Pauli (+3.05) and GTQ (+1.20) show meaningful gaps; cognitive capacity is the most objectively measurable predictor
Competency Execution	0.25	All 10 pillars differ significantly (+1.2 to +2.1); competencies reflect demonstrated behavior in role, not just potential
Work Preferences	0.20	Planning and achievement orientation create the daily behaviors that translate potential into results
Behavioral Strengths	0.15	CliftonStrengths show a clear future-oriented, self-driven theme cluster; DISC diversity means categorical match is secondary
Contextual Fit	0.10	Tenure and grade show minimal differentiation — included for organizational context, not predictive power

Within each TGV, all Talent Variables (TVs) carry equal weight unless overridden via `weights_config`.

Section 2 — SQL Logic & Algorithm

2.1 Approach

The matching algorithm is implemented as a parameterized SQL query using 17 modular CTEs. It is designed to run entirely on-the-fly given a `job_vacancy_id` — no pre-computation, no hardcoded employee IDs.

Script: `sql/matching_algorithm.sql`

2.2 CTE Architecture

```
 CTE 1:  benchmark_employees      → Unpack selected benchmark IDs from
talent_benchmarks
CTE 2:  benchmark_psych          → Psychometric scores for benchmarks
CTE 3:  baseline_psych           → PERCENTILE_CONT(0.5) medians per psych TV
CTE 4:  benchmark_papi           → All 20 PAPI scales pivoted wide for
benchmarks
CTE 5:  baseline_papi            → Median per PAPI scale across benchmarks
CTE 6:  baseline_competency      → Median per pillar (latest year) for
benchmarks
CTE 7:  benchmark_disc_mode      → MODE() DISC type across benchmarks
(categorical baseline)
CTE 8:  benchmark_mbti_mode      → MODE() valid MBTI type across benchmarks
CTE 9:  benchmark_strength_pool  → Pooled unique top-5 CliftonStrengths themes
CTE 10: candidate_info           → All employees + org context (grade, role,
directorate)
CTE 11: candidate_psych          → Psychometric scores per candidate
CTE 12: candidate_papi           → All 20 PAPI scales pivoted wide per candidate
CTE 13: candidate_strengths_overlap → Top-5 theme overlap count per candidate
CTE 14: tv_match_rates_long      → Per-employee × per-TV match rate (long
format)
CTE 15: tgv_match_rates          → AVG of TV match rates within each TGV
CTE 16: tgv_weights              → Hardcoded TGV weights (overridable)
CTE 17: final_match_rates        → Weighted average of TGV rates = Final Match
Rate
```

2.3 Scoring Formulas

Numeric TV (higher is better):

```
LEAST(100, ROUND(user_score / NULLIF(baseline_score, 0) * 100, 2))
```

Numeric TV (lower is better — inverse scales Papi_Z, Papi_K):

```
LEAST(100, GREATEST(0, ROUND(
    (2 * baseline_score - user_score) / NULLIF(baseline_score, 0) * 100, 2
)))
```

Categorical TV (DISC, MBTI — exact match):

```
CASE WHEN user_val = baseline_val THEN 100.0 ELSE 0.0 END
```

→ NULL when either value is missing (no penalty for untested data)

CliftonStrengths overlap:

```
ROUND(overlap_count::NUMERIC / 5 * 100, 2)
-- overlap_count = count of candidate's top-5 themes in benchmark pool
```

TGV match rate:

```
AVG(tv_match_rate)  -- NULLs excluded (missing data is skipped, not penalized)
```

Final match rate:

```
SUM(tgv_match_rate * weight) / SUM(weight)
-- weights: Cognitive 0.30 + Competency 0.25 + Work Prefs 0.20 +
--           Behavioral 0.15 + Contextual 0.10 = 1.00
```

2.4 Sample Output (vacancy with 3 benchmark employees)

The query returns one row per employee × TV combination. Summary view (per employee):

employee_id	fullname	final_match_rate	Cognitive	Competency	Work Prefs	Behavioral	Contextual
EMP101226	Yoga Erlangga Mahendra	92.52%	92.2%	98.0%	86.8%	86.7%	100.0%
EMP101353	Prasetyo Halim	89.85%	88.1%	96.5%	84.2%	80.0%	100.0%
EMP101428	Umar Wibowo	89.05%	90.3%	94.2%	83.1%	80.0%	95.0%

Full output schema:

Column	Description
employee_id	Candidate ID

Column	Description
fullname	Full name
directorate	Organizational unit
role	Position title
grade	Grade level
tgv_name	Talent Group Variable (e.g., Cognitive Ability)
tv_name	Talent Variable (e.g., IQ Score, PAPI Planning)
baseline_score	Benchmark median for this TV
user_score	Candidate score for this TV
tv_match_rate	Match % for this TV (0–100)
tgv_match_rate	Avg/weighted match within TGV
final_match_rate	Weighted overall match %

Section 3 — AI App & Dashboard

3.1 Live Deployment

URL: https://ninditya-talent-match-intelligence.streamlit.app/

Stack: Streamlit 1.40 · Python 3.11 · Supabase (Postgres) · OpenRouter (MiniMax M2.5 free) · Plotly 5.24

3.2 Runtime Inputs

When a manager submits the form, the app: 1. Writes a new `talent_benchmarks` row with a unique `JV-XXXXXXXX` vacancy ID 2. Recomputes baselines dynamically from the selected benchmark employees 3. Runs the Python matching algorithm against all 2,010 employees 4. Calls OpenRouter LLM to generate a structured job profile 5. Renders all visualizations and ranked results

Form fields: - Role Name (text) - Job Level (Entry / Junior / Middle / Senior / Manager / Director) - Role Purpose (1–2 sentences) - Benchmark Employees (multiselect, max 3, Rating=5 only)

3.3 AI-Generated Job Profile

The app calls OpenRouter (MiniMax M2.5 free tier) with a structured prompt that includes: - Role name, level, and purpose - TGV benchmark averages (numeric context for the LLM) - Top CliftonStrengths themes from the benchmark pool

The LLM returns structured JSON:

```
{
  "job_requirements": ["6+ years experience...", "Strong analytical skills..."],
  "job_description": "2-3 sentence narrative...",
  "key_competencies": ["Strategic Thinking", "Data-Driven Decision Making", ...]
}
```

3.4 Dashboard Visualizations

Tab 1 — Job Profile: AI-generated requirements, description, key competencies, benchmark employee list.

Tab 2 — Ranked Talent: Full 2,010-employee ranked table with search and minimum match % filter. Color-gradient on final_match_rate column.

Tab 3 — Dashboard: - Match rate distribution histogram (all 2,010 candidates) - Top-15 candidates horizontal bar chart - TGV heatmap for top-20 candidates (employee × TGV grid)

Tab 4 — Candidate Detail: - TGV radar chart (candidate vs benchmark = 100%) - TV strengths & gaps horizontal bar (all TVs sorted, 80% threshold line) - Full TV-level detail table with color gradient

3.5 Example Flow

```
 Input:  Role = "Head of Digital Marketing"
        Level = Manager
        Purpose = "Drive brand growth and digital acquisition"
        Benchmarks = [EMP100010, EMP100011, EMP100012]

Output: Vacancy ID = JV-D7F4DF9A
        Top match = Yoga Erlangga Mahendra (92.52%)
        AI profile generated in ~8 seconds
        Full ranked table of 2,010 employees
```

Section 4 — Conclusion

What Worked

- The **competency execution signal** is overwhelmingly the strongest predictor of Rating=5 (all 10 pillars $\Delta > +1.2$). Any company seeking to replicate this approach should prioritize behavioral competency assessments over psychometric scores.
- **Pauli (mental arithmetic stamina)** was the best psychometric differentiator — not IQ — which suggests cognitive endurance matters more than peak intelligence for sustained high performance.
- The **counterintuitive PAPI finding** (R5 scores lower on Leadership/G) adds real analytical value: it warns against promoting people based on leadership style alone rather than execution results.

Challenges

- **ERD vs. reality mismatch:** The case study brief mentioned GTQ1–GTQ5 and Tiki1–Tiki4 as separate columns, but the actual database stores single composite columns (`gtq`, `tiki`). Discovering and adapting to this required iterating through the live schema rather than relying on documentation.
- **Model availability:** OpenRouter free-tier models deprecated without notice. Required live API querying to identify a working model (`minimax/minimax-m2.5:free`).
- **Pagination:** Supabase PostgREST caps at 1,000 rows per request. A custom paginated `fetch_table()` function was built to handle the full 2,010-employee dataset.

Ideas for Improvement

1. **Longitudinal weighting:** Use 5-year competency trend (slope) instead of single latest-year score — an employee improving rapidly may be a better bet than one plateauing at a high level.
2. **Role-specific Success Formulas:** Train separate weight sets per job family (e.g., commercial roles weight Papi_P higher; technical roles weight GTQ and Cognitive higher).
3. **Ensemble baseline:** Replace median with a weighted blend of the 3 benchmark employees based on their performance trajectory, not just latest rating.
4. **Interpretable gap analysis:** Auto-generate a personalized development plan for each candidate highlighting their top 3 gaps and suggested interventions.
5. **Real-time retraining:** As new performance ratings arrive annually, automatically re-run the EDA and update the Success Formula weights.

Analysis conducted on Supabase (Postgres) · Python 3.11 · Pandas · Plotly · Streamlit Repository: https://github.com/ninditya/talent-match-intelligence App: https://ninditya-talent-match-intelligence.streamlit.app/